

Chebyshev Cantelli Inequality

$$S_k = \{ i \mid x_i - \bar{x} \geq ks \}$$

std. dev.

$$\text{To prove: } \frac{|S_k|}{n} \leq \frac{1}{1+k^2}$$

Proof:

$$y_i := x_i - \bar{x} \quad \triangleq$$

Assume $b > 0$.

$$\sum_{i=1}^n (y_i + b)^2 \geq \sum_{i: y_i \geq ks} (y_i + b)^2$$

$$\geq \sum_{i: y_i \geq ks} (ks + b)^2$$

$$= |S_k| (ks + b)^2 \rightarrow \textcircled{1}$$

$$\sum_{i=1}^n (y_i + b)^2 = \sum_{i=1}^n y_i^2 + \left(\sum_{i=1}^n y_i \right) 2b + nb^2 \quad \xrightarrow{0}$$

$$= \sum_{i=1}^n (x_i - \bar{x})^2 + nb^2$$

$$= (n-1)s^2 + nb^2 \rightarrow \textcircled{2}$$

$$|S_k| (ks + b)^2 \leq (n-1)s^2 + nb^2$$

$$\begin{aligned} \therefore \frac{|S_k|}{n} &\leq \frac{s^2 - \frac{1}{n}s^2 + b^2}{(ks + b)^2} \\ &\leq \frac{s^2 + b^2}{(ks + b)^2} \end{aligned}$$

$$J(b) = \frac{s^2 + b^2}{(ks + b)^2}$$

$$\frac{(ks+b)^2 \cancel{2b} - (s^2+b^2) \cancel{2}(ks+b)}{(ks+b)^4} = 0$$

$$(ks+b)^{\cancel{2}} b = (\cancel{ks+b}) (s^2 + b^2)$$

$$ksb + b^2 = s^2 + b^2$$

$$b = s/k$$

$$\frac{|S_k|}{n} \leq \frac{s^2 + s^2/k^2}{(ks + s/k)^2} = \frac{1 + 1/k^2}{(k + 1/k)^2} = \frac{1}{k^2 + 1}$$